

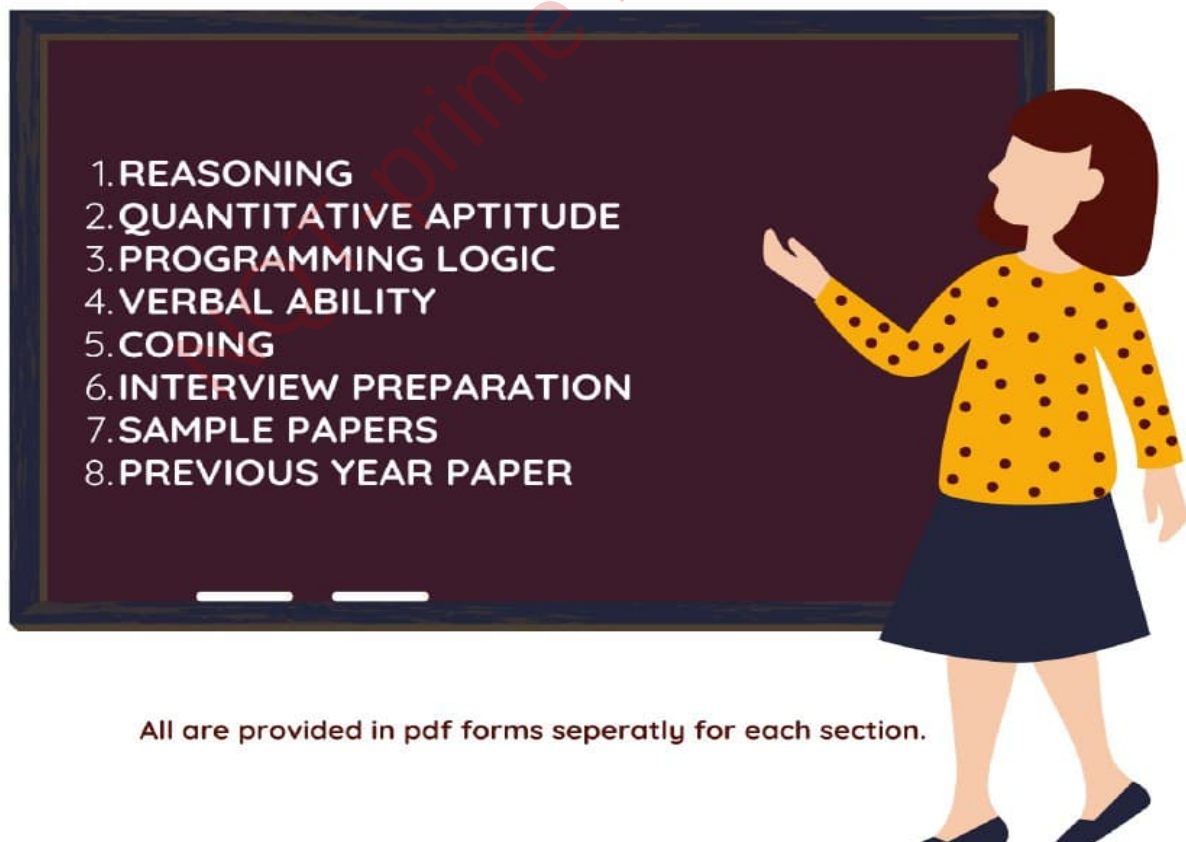
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TCS NQT PREMIUM

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All are provided in pdf forms seperatly for each section.

Q.A pointer variable can be

- 1. Changed within function.**
- 2. Assigned an integer value.**
- 3. None of these**
- 4. Passed to a function as argument.**

Correct Op: 4

Q. Which of the following uses structure?

- 1. Linked Lists**
- 2. Array of structures**
- 3. All of these**
- 4. Binary Tree**

Correct Op: 3

Q. Strings are character arrays. The last index of it contains the null-terminated character

-
- 1. \t**
 - 2. \1**
 - 3. \0**
 - 4. \n**

Correct Op: 3

Q. Which of the following is a collection of different data types?

- 1. String**
- 2. Structure**
- 3. Array**

4. Files Correct Op:2

Q. What function should be used to free the memory allocated by calloc() ?

- 1. free();**
- 2. malloc(variable_name,0)**
- 3. dealloc();**

4. memalloc(variable_name, 0) Correct Op:1

Q. In the standard library of C programming language, which of the following header file is designed for basic mathematical operations?

- 1. conio.h**
- 2. stdio.h**
- 3. math.h**
- 4. Dos.h Correct Op:3**

Q. int **ptr; is?

- 1. Pointer to integer**
- 2. None of these**
- 3. Pointer to pointer**
- 4. Invalid declaration Correct Op: 3**

Q8. Which of the following special symbol allowed in a variable name?

1. (underscore)
2. -(hyphen)
3. |(pipeline)
4. * (asterisk) Correct Op: 1

Q9. All keywords in C are in

1. Uppercase letters
2. None of these
3. Lowercase letters
4. Camel Case letters

Correct Op:3

Q10. What should the program below print?

```
#include<stdio.h> #include<string.h>
```

```
#include<stdlib.h>
void myfunc(char** param){
    ++param;
}
int main(){
    char* string = (char*)malloc(64); strcpy(string, "hello_World"); myfunc(&string);
    myfunc(&string); printf("%s\n", string);
    // ignore memory leak for sake of quiz return 0;
}
```

1. hello_World
2. ello_World
3. lo_World
4. llo_World

Correct Op:1

Q: What is the output of this C code? #include <stdio.h>

```
void main()
{

    int k = 5;
    int*p=&k;
    int **m = &p; printf("%d%d%d\n", k, *p, **p); }
a) 5 5 5
b) 55junk
```

- c) 5 junk junk
- d) Compile timeerror

Correct op: D

Explanations

1) It would have been 5 5 5 if it were **m and not**p.

Q. Which of the following statements about stdout and stderr are true? a) They both are the same

b) Run time errors are automatically displayed instderr

c) Both are connected to the screen by default.

d) stdout is line buffered but stderr is unbuffered.

Correct Op: D

Explanation -

a) False. b) Not by default. c) Not by default. d) True.

Q: Given the below statements about C programming language:

1) main() function should always be the first function present in a C program file 2) all the elements of an union share their memory location

3) A void pointer can hold address of any type and can be typecasted to any type 4) A static variable hold random junk value if it is not initialised

Which of the above are correct statements? A) 2,3

B) 1,2

C) 1,2,3

D) 1,2,3,4

Correct Op - A

Explanations

In a file you can write a function before main() - False

all the elements of an union share their memory location - True.

A void pointer can hold address of any type and can be typecasted to any type - True Static value - False as value is 0

In C, if an object that has static storage duration is not initialized explicitly, then:

- if it has pointer type, it is initialized to a NULL pointer;
- if it has arithmetic type, it is initialized to (positive or unsigned) zero;
- if it is an aggregate, every member is initialized (recursively) according to these rules;
- if it is a union, the first named member is initialized (recursively) according to these rules.

Q If a function is defined as static, it means

- A)** The value returned by the function does not change
- B)** all the variable declared inside the function automatically will be assigned initial value of zero
- C)** It should be called only within the same source code / program file.
- D)** None of the other choices as it is wrong to add static prefix to a function

Correct Op: C

Access to static functions is restricted to the file where they are declared. Therefore, when we want to restrict access to functions, we make them static.

Q: Comment on the below while statement=

while (0 == 0) { }

- A)** It has syntax error as there are no statements within braces { }
- B)** It will run forever
- C)** It compares 0 with 0 and since they are equal it will exit the loop immediately
- D)** It has syntax error as the same number is being compared with itself

Correct Op: B

while(0==0) { } is equivalent to **while(1) { }**

1. What will happen if in a C program you assign a value to an array element whose subscript exceeds the size of array?

- A.** The element will be set to 0.
- B.** The compiler would report an error.
- C.** The program may crash if some important data gets overwritten.
- D.** The array size would appropriately grow.

Answer: Option C

Explanation:

If the index of the array size is exceeded, the program will crash. Hence "option c" is the correct answer. But the modern compilers will take care of this kind of errors.

2. What does the following declaration mean? **int(*ptr)[10];**

- A.** ptr is array of pointers to 10 integers
- B.** ptr is a pointer to an array of 10 integers
- C.** ptr

is an array of 10 integers

- D.** ptr is an pointer to array

Answer: Option B

3. In C, if you pass an array as an argument to a function, what actually gets passed?

- A. Value of elements in array
- B. First element of the array
- C. Base address of the array

D. Address of the last element of array Answer: Option C

Explanation:

The statement 'C' is correct. When we pass an array as a function argument, the base address of the array will be passed.

4. What will be the output of the program ? #include <stdio.h>

```
int main()
{
```

```
    int a[5] = {5, 1, 15, 20, 25}; int i, j, m;
    i = ++a[1];
    j = a[1]++;
```

```
    m = a[i++];
    printf("%d, %d, %d", i, j, m); return 0;
```

```
}
```

A. 2, 1, 15

B. 1, 2, 5

C. 3, 2, 15

D. 2, 3, 20

Answer: Option C

Explanation:

Step 1: `int a[5] = {5, 1, 15, 20, 25};` The variable arr is declared as an integer array with a size of 5 and it is initialized to

`a[0] = 5, a[1] = 1, a[2] = 15, a[3] = 20, a[4] = 25 .`

Step 2: `int i, j, m;` The variable i, j, m are declared as an integer type. Step 3: `i = ++a[1];` becomes `i = ++1;` Hence `i = 2` and `a[1] = 2`

Step 4: `j = a[1]++;` becomes `j = 2++;` Hence `j = 2` and `a[1] = 3`.

Step 5: `m = a[i++]`; becomes `m = a[2]`; Hence `m = 15` and `i` is incremented by 1 (`i++` means `2++` so `i=3`)

Step 6: `printf("%d, %d, %d", i, j, m)`; It prints the value of the variables `i, j, m` Hence the output of the program is 3, 2, 15

5. Is there any difference in the following declarations? `int fun(int arr[]);` `int fun(int arr[2]);`

A. Yes

B. No

Answer: Option B

Explanation:

No, both the statements are same. It is the prototype for the function `fun()` that accepts one integer array as an parameter and returns an integer value.

6. Are the expressions `sarr` and `&arr` same for an array of 10 integers? A. Yes B. No

Answer: Option B

Explanation:

Both mean two different things. `arr` gives the address of the first int, whereas the `&arr` gives the address of array of ints.

7. Which of the following statements should be used to obtain a remainder after dividing 3.14 by 2.1?

A. `rem = 3.14 % 2.1;`

B. `rem = modf(3.14, 2.1);`

C. `rem = fmod(3.14, 2.1);`

D. Remainder cannot be obtained in floating point division.

Answer: Option C

Explanation:

`fmod(x, y)` - Calculates `x` modulo `y`, the remainder of `x/y`.

This function is the same as the modulus operator. But fmod() performs floating point divisions.

8. What are the types of packages? A. Internal and External B. External, Internal and None C. External and None

D. Internal Answer:

Option B

Explanation:

External package -> means global, non-static variables and functions. Internal package -> means static variables and functions with file scope. None package -> means Local variable

9. Which of the following special symbols are allowed in a variable name? A. * (asterisk) B. | (pipe)

C. - (hyphen)

D. _ (underscore)

Answer: Option D

Explanation:

Variable names in C are made up of letters (upper and lower case) and digits. The underscore character ("_") is also permitted. Names must not begin with a digit.

Examples of valid (but not very descriptive) C variable names: => foo

=> Bar

=> BAZ

=> foo_bar => _foo42 => _

=> QuUx

10. Is there any difference between the following declarations? 1 : extern int fun();

2 : int fun();

A. Both are identical

B. No difference, except extern int fun(); is probably in another file C. int fun(); is overridden with extern int fun();

D. None of these

Answer: Option B

Explanation:

`extern int fun();` declaration in C is to indicate the existence of a global function and it is defined externally to the current module or in another file.

`int fun();` declaration in C is to indicate the existence of a function inside the current module or in the same file.

Question 1: Use of an increment statement or decrement statement in C? Answer:

There are actually two ways you can do this. One is to use the increment operator `++` and decrement operator `--`. For example, the statement `x++` means to increment the value of `x` by 1. Likewise, the statement `x--` means to decrement the value of `x` by 1.

Two types of increments are:

1. pre increment: (increment by 1 then print)and

2. post increment: (print then incremented value will be in buffer). Same thing will be with decrement.

Question 2: In programs we place comment symbols on some codes instead of deleting it. How does this aid in debugging?

Answer:

Placing comment symbols `/* */` around a code, also referred to as commenting out, is a way of isolating some codes that you think maybe causing errors in the program, without deleting the code.

Question 3: What is the use of a '\0' character? Answer:

This character is used primarily to show the end of a string value.

Question 4: What is the difference between the = symbol and == symbol? Answer:

The `=` symbol is often used in mathematical operations. It is used to assign a value to a given variable. On the other hand, the `==` symbol, also known as equal to or equivalent to, is a relational operator that is used to compare two values.

Question 5: In C Programming, which of the following operators is incorrect and why? (`>=`, `<=`, `<>`, `==`)

Answer:

`<>` is incorrect, all other operators are relational operators. While this operator is correctly interpreted as not equal to in writing conditional statements, it is not the proper operator to be used in C programming. Instead, the operator `!=` must be used to indicate not equal to condition.

Question 6: Can the curly brackets { } be used to enclose a single line of code? Answer:

While curly brackets are mainly used to group several lines of codes, it will still work without error if you used it for a single line. Some programmers prefer this method as a way of organizing codes to make it look clearer, especially in conditional statements.

Question 7: Can I use int data type to store the value 32768? Why/why not? Answer:

No. int data type is capable of storing values from -32768 to 32767. To store 32768, you can use long int instead. You can also use 'unsigned int, assuming you don't intend to store negative values.

Question 8: Can two or more operators such as \n and \t be combined in a single line of program code?

Answer: Yes, it's perfectly valid to combine operators, especially if the need arises.

For example: you can have a code like 'printf ('Hello\n\n'World\')' to output the text 'Hello' on the first line and 'World' enclosed in single quotes to appear on the next two lines.

Question 9: When is the 'void' keyword used in a function? Answer:

When declaring functions, you will decide whether that function would be returning a value or not. If that function will not return a value, such as when the purpose of a function is to display some outputs on the screen, then void is to be placed at the leftmost part of the function header. When a return value is expected after the function execution, the data type of the return value is placed instead of void.

Question 10: Write a loop statement that will show the following output: 1 12

123
1234

12345 **Answer:**

```
for (a=1; a<=5; i++) { for (b=1; b<=a; b++) printf("%d",b);
```

```
printf("\n");  
}
```

Question 1: How would you round off a value from 1.66 to 2.0?

- A. ceil(1.66)
- B. floor (1.66)
- C. roundup (1.66) D. Roundto(1.66)

Answer: A

```
/* Example for ceil() and floor() functions: */ #include<stdio.h>
#include<math.h>
int main()

{
printf("\n Result : %f" , ceil(1.44) ); printf("\n Result : %f" , ceil(1.66) ); printf("\n
Result : %f" , floor(1.44) ); printf("\n Result : %f" , floor(1.66) ); return 0;

}
// Output:
// Result : 2.000000

// Result : 2.000000 // Result : 1.000000 // Result : 1.000000
```

Question 2: What will be the output of the program? #include<stdio.h>
intX=40;
intmain()

```
{
int X=20; printf("%d\n", X); return 0;

}
```

A.20

B.40

C.Error

D.No Output

Answer:A

Whenever there is conflict between a local variable and global variable, the local variable gets priority.

Question 3: A long double can be used if range of a double is not enough to accommodate a real number.

A. True

B. False

Answer: A

True, we can use long double; if double range is not enough. Double = 8 bytes.

Long double = 10 bytes.

Question 4: A float is 4 bytes wide, whereas a double is 8 bytes wide. A. True B. False

Answer: A

True,

float = 4 bytes. Double = 8 bytes.

Question 5: If the definition of the external variable occurs in the source file before its use in a particular function, then there is no need for an extern declaration in the function.

A. True

B. False Answer: A

True, when a function is declared inside the source file, that function (local function) get a priority than the extern function. So there is no need to declare a function as extern inside the same source file

Question 6: If the definition of the external variable occurs in the source file before its use in a particular function, then there is no need for an extern declaration in the function.

A. True

B. False Answer: A

True, When a function is declared inside the source file, that function (local function) get a priority than the extern function. So there is no need to declare a function as extern inside the same source file

Question 7: Size of short integer and long integer can be verified using the sizeof() operator.

A. True

B. False

Answer: A

True, we can find the size of short integer and long integer using the sizeof() operator.

Question 8: Range of double is $-1.7e-38$ to $1.7e+38$ (in 16 bit platform - Turbo C under DOS)

A. True

B. False

Answer: B

False, the range of double is $-1.7e-308$ to $1.7e+308$.

Question 9: Size of short integer and long integer would vary from one platform to another.

A. True

B. False

Answer:A

True, Depending on the operating system/compiler/system architecture you are working on, the range of data types can vary.

Question 10: Range of float is $-2.25e-308$ to $2.25e+308$ A. True

B. False

Answer: Option B

False, the range of float is $-3.4e-38$ to $3.4e+38$.

Question 1: What is wrong in this statement? `scanf(%d,whatnumber);`

Answer:

An ampersand '&' symbol must be placed before the variable name whatnumber. Placing & means whatever integer value is entered by the user is stored at the address of the variable name. This is a common mistake for programmers, often leading to logical errors.

Question2:Whatdoestheformat%10.2meanwhenincludedinaprintfstatement?

Answer:

This format is used for two things: to set the number of spaces allotted for the output number and to set the number of decimal places. The number before the decimal point is for the allotted space, in this case it would allot 10 spaces for the output number. If the number of space occupied by the output number is less than 10, addition space

characterswillbeinsertedbeforetheactualoutputnumber.Thenumberafterthedecimalpointsetsthenumberofdecimal places,in thiscase,it's2decimal spaces.

Question 3: What are linked list?

Answer:

A linked list is composed of nodes that are connected with another. In C programming, linked lists are created using pointers. Using linked lists is one efficient way of utilizing memory for storage.

Question 4: What are binary trees? Answer:

Binary trees are actually an extension of the concept of linked lists. A binary tree has two pointers, a left one and a right one. Each side can further branch to form additional nodes, which each node having two pointers as well.

Question 5: Differences between C and Java? Answer:

JAVA is Object-Oriented while C is procedural.

2. Java is an Interpreted language while C is a compiled language.

3. C is a low-level language while JAVA is a high-level language.

4. C uses the top-down approach while JAVA uses the bottom-up approach.

5. Pointer goes backstage in JAVA while C requires explicit handling of pointers.

Question 6: In header files whether functions are declared or defined?

Answer: Functions are declared within header file. That is function prototypes exist in a header file, not function bodies. They are defined in library (lib).

Question 7: What are the different storage classes in C?

Answer:

There are four types of storage classes in C. They are extern, register, auto and static.

Question 8: What does static variable mean? Answer:

Static is an access qualifier. If a variable is declared as static inside a function, the scope is limited to the function, but it will exist for the life time of the program. Values will be persisted between successive calls to a function.

Question 9: How do you print an address? Answer:

Use %p in printf to print the address.

Question 10: What are macros? What are its advantages and disadvantages?

Answer:

Macros are processor directive which will be replaced at compile time.

The disadvantage with macros is that they just replace the code they are not function calls. Similarly the advantage is they can reduce time for replacing the same values.

Question 1: Difference between pass by reference and pass by value? Answer:

Pass by value just passes the value from caller to calling function so the called function cannot modify the values in caller function. But Pass by reference will pass the address to the caller function instead of value. If called function requires to modify any value it can directly modify.

Question 2: What is an object? Answer:

Object is a software bundle of variables and related methods. Objects have state and behaviour.

Question 3: What is a class? Answer:

Class is a user-defined data type in C++. It can be created to solve a particular kind of problem. After creation the user need not know the specifics of the working of a class.

Question 4: What is the difference between class and structure? Answer:

Structure: Initially (in C) a structure was used to bundle different type of data types together to perform a particular functionality. But C++ extended the structure to contain functions also.

The major difference is that all declarations inside a structure are by default public. Class: Class is a successor of Structure. By default all the members inside the class are private.

Question 5: What is pointer? Answer:

Pointer is a variable in a program is something with a name, the value of which can vary. The way the compiler and linker handles this is that it assigns

a specific block of memory within the computer to hold the value of that variable.

Question 6: What is the difference between null and void pointer? Answer:

A Null pointer has the value 0. Void pointer is a generic pointer introduced by ANSI. Generic pointer can hold the address of any data type.

Question 7: what is function overloading? Answer:

Function overloading is a feature of C++ that allows us to create multiple functions with the same name, so long as they have different parameters. Consider the following function:

```
int Add(int nX, int nY) {  
  
    return nX + nY; }
```


Question 8: what is friend function? Answer:

A friend function for a class is used in object-oriented programming to allow access to public, private, or protected data in the class from the outside.

Normally, a function that is not a member of a class cannot access such information; neither can an external class. Occasionally, such access will be advantageous for the programmer. Under these circumstances, the function or external class can be declared as a friend of the class using the friend keyword

Question 9: What do you mean by inline function?

Answer: The idea behind inline functions is to insert the code of a called function at the point where the function is called. If done carefully, this can improve the application's performance in exchange for increased compile time and possibly (but not always) an increase in the size of the generated binary executables.

Question 10: Tell me something about abstract classes? Answer:

An abstract class is a class which does not fully represent an object. Instead, it represents a broad range of different classes of objects. However, this representation extends only to the features that those classes of objects have in common. Thus, an abstract class provides only a partial description of its objects.

Question 1: What is the difference between an array and a list? Answer:

Array is collection of homogeneous elements. List is collection of heterogeneous elements.

For Array memory allocated is static and continuous. For List memory allocated is dynamic and random.

Array: User need not have to keep in track of next memory allocation.

List: User has to keep in Track of next location where memory is allocated.

Array uses direct access of stored members; list uses sequential access for members.

Question 2: What are the differences between structures and arrays? Answer:

Arrays are a group of similar data types but Structures can be a group of different data types.

Question 3: What is data structure? Answer:

A data structure is a way of organizing data that considers not only the items stored, but also their relationship to each other. Advance knowledge about the relationship between data items allows designing of efficient algorithms for the manipulation of data.

Question 4: Can you list out the areas in which data structures are applied extensively?

Answer:

**Compiler Design,
Operating System,**

**Database Management System, Statistical analysis package, Numerical Analysis,
Graphics,**

Question 5: What are the advantages of inheritance? Answer:

It permits code reusability. Reusability saves time in program development. It encourages the reuse of proven and debugged high-quality software, thus reducing problem after a system becomes functional.

Question 6: Advantages of a macro over a function? Answer:

Macro gets to see the Compilation environment, so it can expand #defines. It is expanded by the pre-processor.

Question 7: What is command line argument? Answer:

Getting the arguments from command prompt in c is known as command line arguments. In c main function has three arguments. They are:

Argument counter Argument vector Environment vector

Question 8: What are the 4 basics of OOP?

Answer:

Abstraction, Inheritance, Encapsulation, and Polymorphism.

Question 9: Tell how to check whether a linked list is circular. Answer:

Create two pointers, each set to the start of the list. Update each as follows: while

```
(pointer1) {  
pointer1 = pointer1->next; pointer2=pointer2-  
>next;if(pointer2)pointer2=pointer2->next; if  
(pointer1 == pointer2) {  
print ("circular\n");
```

```
}}
```

Question 10: Write a program to swap two numbers without using a temporary variable.

Answer:

```
void swap(int &i, int &j)  
{  
  
i=i+j ; j=i- j; i=i-j; }
```

Ques. 1 Which is the character array used to accept command line arguments?

- A) charargv
- B) char*argv[]
- C) charargv[]
- D) char*argv

Ques. 2 What is a dangling pointer?

Ques. 3 Which is not a string function?

- A) strstr
- B) strcmp
- C)strupr
- D) strchr

Ques. 4 Which of the following does not require to include math.h header file?

- A) pow()

- B) rand()
- C) sqrt()
- D) sinh()

Ques. 5 What is the task of pre-processor? A) Expanding

- B) Compiling
- C) Linking
- D) All of the above

Ques. 6 Which of the following is true?

- A) realloc() can change the memory size of arrays
- B) Unary operator works on only one operand
- C) Struct and Union works in same way.
- D) None of the above

Ques. 7 Which of this is used to skip one iteration:

- A) break
- B) continue
- C) goto
- D) return

Ques. 8 Which address does a pointer to an array store:

- A) Memory address of the first element of the array Don't remember the other options.

Ques. 9 Predict the output:

```
float a = 0.1;  
if(a==0.1)  
printf("Yes");
```

```
else  
printf("No"); Answer would be No.
```

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Ques. 10 Another output based question which basically displayed the input string in reverse pattern.

For example, ABRACADABRA was displayed as ARBADACARBA.

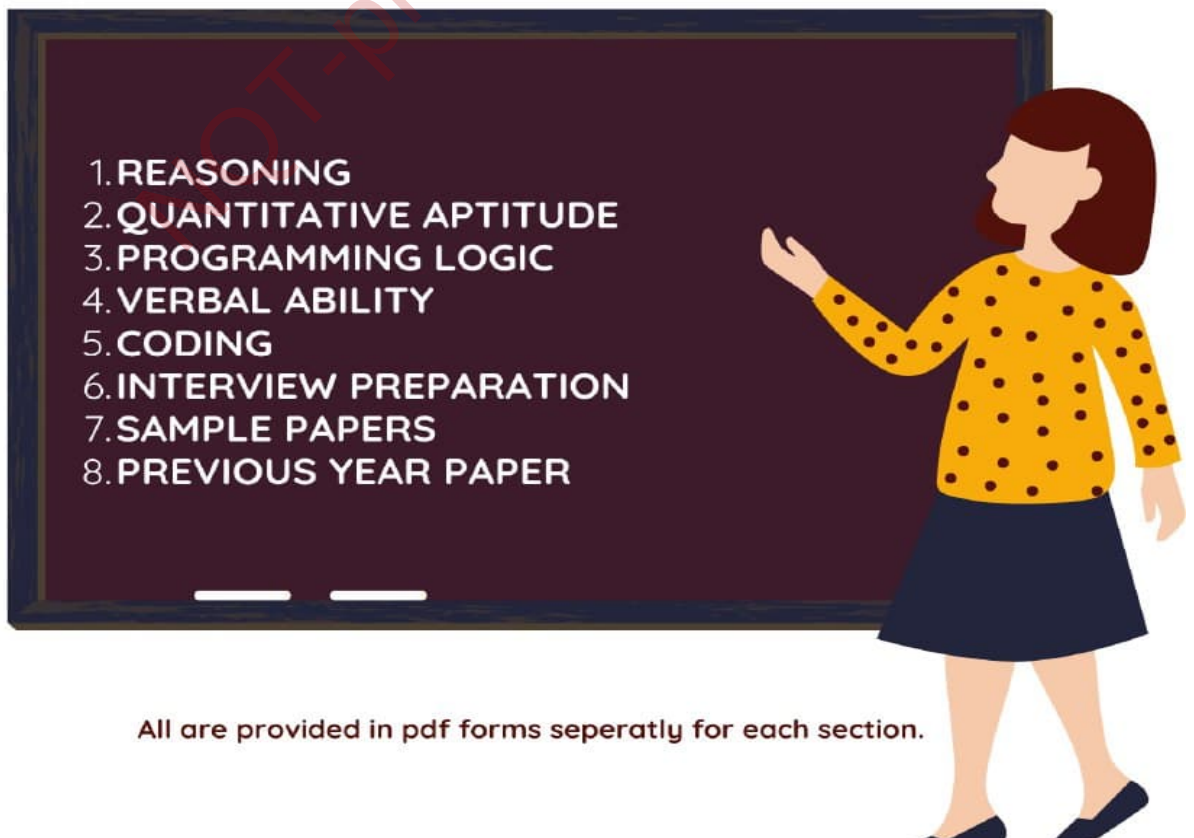
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